Fiscal Year:	FY 2007	**Task Last Updated:**	FY 11/16/2007
PI Name:	Bunick, Gerard J. Ph.D.		
Project Title:	Structural Studies of Nucleosomes and Chromatin		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gjbunick@utk.edu	**Fax:**	FY 865-974-7917
PI Organization Type:	UNIVERSITY	**Phone:**	865-974-6488
Organization Name:	University of Tennessee		
PI Address 1:	UT/ORNL Graduate School of Genome Science and Technology		
PI Address 2:	1060 Commerce Park		
PI Web Page:			
City:	Oak Ridge	**State:**	TN
Zip Code:	37830-8026	**Congressional District:**	3
Comments:	Deceased September 2007.		
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	10/01/2001	**End Date:**	11/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Miller, Teresa	**Contact Phone:**	256-544-7815
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG8-1826		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This research program is exploring the production of large nucleosome core particle crystals suitable for neutron diffraction to examine the role of hydration in chromatin, and to incorporate H3-replacing centromeric proteins in the histone octamer and reconstituted nucleosome core particles (NCP). Neutron structures of D-xylose isomerase are being determined and analyzed to reveal the protonation states of active-site amino acid residues and water molecules. The first task is to conduct research to incorporate H3-replacing centromeric proteins in the histone octamer and reconstituted nucleosome core particles (NCP). The structures will provide additional critical insights into structure-function relationships in chromatin. The second task will elucidate the mechanism of D-xylose isomerase using information obtained from neutron diffraction studies. Structures of modified nucleosome core particles are important to the elucidation of the molecular processes of gene		

	regulation and transcription in chromatin, the substrate for all such activity in humans. The knowledge acquired from this research may provide answers to the genesis and proliferation of cancer and other diseases. The studies of D-xylose isomerase demonstrate the elucidation of enzyme mechanisms by combining neutron diffraction with traditional X-ray structure determination.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Structural studies are demonstrating the power of combining X-ray diffraction results with spallation neutron diffraction for the elucidation of enzyme mechanisms. Our neutron results may encourage the construction of new macromolecular diffraction instruments at spallation sources now under construction in the United States and abroad. The proposed research will extend our knowledge of chromatin structural biology, providing information linking the structure of chromatin to the regulation of gene expression and control. The results may help to understand aspects of cancer and other diseases.
Task Progress:	PI deceased as of September 2007. Peer-reviewed journal articles resulting from his research are listed in the Bibliography section.
Bibliography Type:	Description: (Last Updated: 11/16/2007)
Articles in Peer-reviewed Journals	Hanson BL, Bunick GJ. "Annealing macromolecular crystals." Methods Mol Biol. 2007;364:31-42. PMID: 17172759 , Feb-2007
Articles in Peer-reviewed Journals	Katz AK, Li X, Carrell HL, Hanson BL, Langan P, Coates L, Schoenborn BP, Glusker JP, Bunick GJ. "Locating active-site hydrogen atoms in D-xylose isomerase: time-of-flight neutron diffraction." Proc Natl Acad Sci U S A. 2006 May 30;103(22):8342-7. PMID: 16707576 , May-2006
Articles in Peer-reviewed Journals	Young GW, Hoofring SA, Mamula MJ, Doyle HA, Bunick GJ, Hu Y, Aswad DW. "Protein L-isoaspartyl methyltransferase catalyzes in vivo racemization of Aspartate-25 in mammalian histone H2B." J Biol Chem. 2005 Jul 15;280(28):26094-8. PMID: 15908425 , Jul-2005
Articles in Peer-reviewed Journals	Bunick CG, Nelson MR, Mangahas S, Hunter MJ, Sheehan JH, Mizoue LS, Bunick GJ, Chazin WJ. "Designing sequence to control protein function in an EF-hand protein." J Am Chem Soc. 2004 May 19;126(19):5990-8. PMID: 15137763 , May-2004
Articles in Peer-reviewed Journals	Hanson BL, Alexander C, Harp JM, Bunick GJ. "Preparation and crystallization of nucleosome core particle." Methods Enzymol. 2004;375:44-62. PMID: 14870658 , Mar-2004
Articles in Peer-reviewed Journals	Hanson BL, Langan P, Katz AK, Li X, Harp JM, Glusker JP, Schoenborn BP, Bunick GJ. "A preliminary time-of-flight neutron diffraction study of Streptomyces rubiginosus D-xylose isomerase." Acta Crystallogr D Biol Crystallogr. 2004 Feb;60(Pt 2):241-9. PMID: 14747699 , Feb-2004
Articles in Peer-reviewed Journals	Hanson BL, Harp JM, Bunick GJ. "The well-tempered protein crystal: annealing macromolecular crystals." Methods Enzymol. 2003;368:217-35. PMID: 14674276 , Jan-2004
Articles in Peer-reviewed Journals	Hanson BL, Schall CA, Bunick GJ. "New techniques in macromolecular cryocrystallography: macromolecular crystal annealing and cryogenic helium." J Struct Biol. 2003 Apr;142(1):77-87. Review. PMID: 12718921 , Jan-2004
Articles in Peer-reviewed Journals	Schurter BT, Koh SS, Chen D, Bunick GJ, Harp JM, Hanson BL, Henschen-Edman A, Mackay DR, Stallcup MR, Aswad DW. "Methylation of histone H3 by coactivator-associated arginine methyltransferase 1." Biochemistry. 2001 May 15;40(19):5747-56. PMID: 11341840 , Jan-2004